

EXPERIMENT: DATA VISUALIZATION USING EXCEL

AIM: To create simple interactive excel dashboard for given data.

Outcomes: Students are able to understand visualization importance and create simple dashboard using Excel.

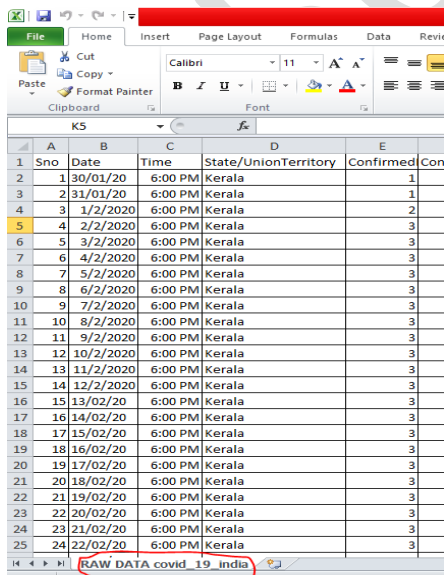
Apparatus/Software required: data sheet in .xls or CSV format, Excel 10, System/laptop with minimum specifications.

Theory: Data visualization refers to techniques used to communicate insights from data through visual representation. Its main goal is to display large datasets into visual graphics to allow for easy understanding of complex relationships within the data. It is often used interchangeably with terms such as information graphics, statistical graphics, and information visualization.

Because of the way the human brain processes information, using charts or graphs to visualize large amounts of complex data is easier than poring over spreadsheets or reports. Data visualization is a quick, easy way to convey concepts in a universal manner and you can experiment with different scenarios by making slight adjustments.

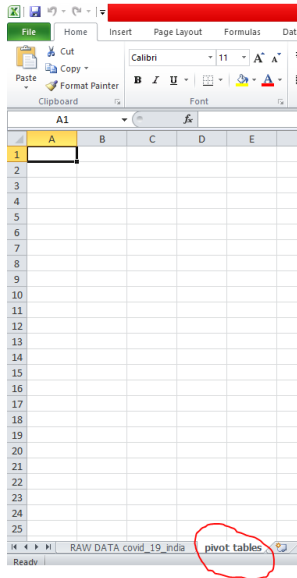
Procedure:

- 1) Check the data from sheet and clean the data whichever not essential to make dashboard.
- 2) Rename the sheet as RAW data as shown below.

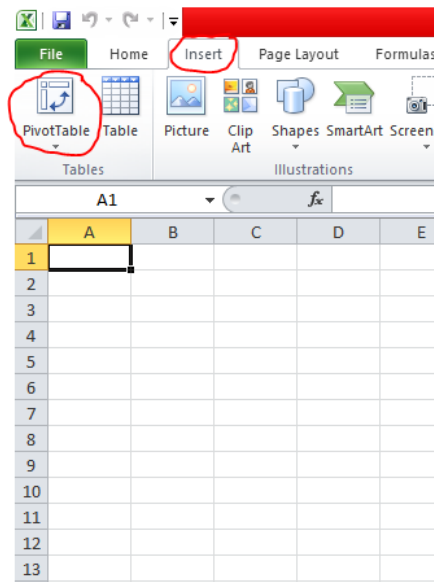


Sno	Date	Time	State/Union Territory	Confirmed	Con
1	30/01/20	6:00 PM	Kerala		1
2	31/01/20	6:00 PM	Kerala		1
3	1/2/2020	6:00 PM	Kerala		2
4	2/2/2020	6:00 PM	Kerala		3
5	3/2/2020	6:00 PM	Kerala		3
6	4/2/2020	6:00 PM	Kerala		3
7	5/2/2020	6:00 PM	Kerala		3
8	6/2/2020	6:00 PM	Kerala		3
9	7/2/2020	6:00 PM	Kerala		3
10	8/2/2020	6:00 PM	Kerala		3
11	9/2/2020	6:00 PM	Kerala		3
12	10/2/2020	6:00 PM	Kerala		3
13	11/2/2020	6:00 PM	Kerala		3
14	12/2/2020	6:00 PM	Kerala		3
15	13/02/20	6:00 PM	Kerala		3
16	14/02/20	6:00 PM	Kerala		3
17	15/02/20	6:00 PM	Kerala		3
18	16/02/20	6:00 PM	Kerala		3
19	17/02/20	6:00 PM	Kerala		3
20	18/02/20	6:00 PM	Kerala		3
21	19/02/20	6:00 PM	Kerala		3
22	20/02/20	6:00 PM	Kerala		3
23	21/02/20	6:00 PM	Kerala		3
24	22/02/20	6:00 PM	Kerala		3

3) Create new sheet and rename it as pivot tables.

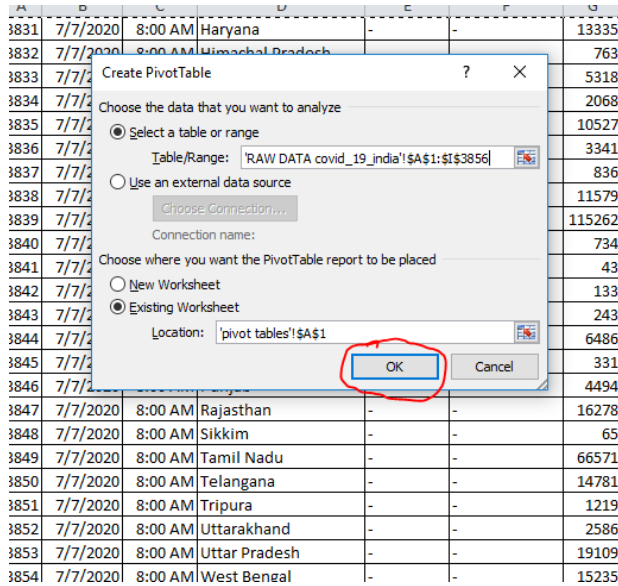


4. in pivot table sheet click on insert option and then click on pivot table.

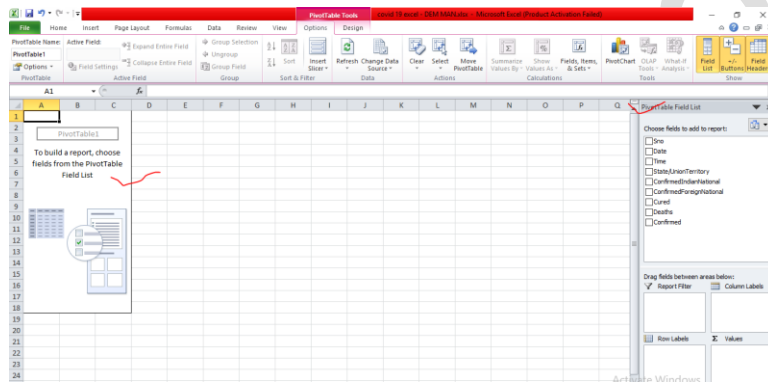


5. Now you got pivot table asking range of the sheet or data for which table to be put. Go to raw data sheet and select the range and click ok.

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6) Now the pivot table is ready with pivot table field list.



7) Select data variable for which you would like to put chart and drag them to appropriate areas mention at the bottom of PTF list (Drag and drop). Example states in ROW LABEL and count of cured value in VALUES. Now segregated data has been filtered from raw sheet and generated separate table.

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STATE	Sum of CURED
Andaman and Nicobar Islands	74
Andhra Pradesh	8920
Arunachal Pradesh	92
Assam	7882
Bihar	8997
Chandigarh	401
Chhattisgarh	2667
Dadar Nagar Haveli	2
Daman & Diu	0
Delhi	72088
Goa	1061
Gujarat	26315
Haryana	13335
Himachal Pradesh	763
Jammu and Kashmir	5318
Jharkhand	2068
Karnataka	10527
Kerala	3341
Ladakh	836
Madhya Pradesh	11579
Maharashtra	115262
Manipur	734
Meghalaya	43
Mizoram	133

8) Repeat the procedure and create multiple pivot tables as per requirement on same sheet. In this examples three pivot tables are formed a) state wise cured count b) state wise death and c) state wise cured, death, confirmed etc.

STATE	Sum of CURED	Sum of DEATH	Sum of CURED	Sum of DEATH	Sum of CONFIRMED
Andaman and Nicobar Islands	74	0	74	0	141
Andhra Pradesh	8920	239	8920	239	20019
Arunachal Pradesh	92	2	92	2	270
Assam	7882	14	7882	14	12160
Bihar	8997	97	8997	97	12125
Chandigarh	401	6	401	6	489
Chhattisgarh	2667	14	2667	14	3305
Dadar Nagar Haveli	2	0	2	0	26
Daman & Diu	0	0	0	0	2
Delhi	72088	3115	72088	3115	100823
Goa	1061	7	1061	7	1813
Gujarat	26315	1960	26315	1960	36772
Haryana	13335	276	13335	276	17504
Himachal Pradesh	763	11	763	11	1077
Jammu and Kashmir	5318	138	5318	138	8675
Jharkhand	2068	20	2068	20	2847
Karnataka	10527	401	10527	401	25317
Kerala	3341	27	3341	27	5622
Ladakh	836	1	836	1	1005
Madhya Pradesh	11579	617	11579	617	15284
Maharashtra	115262	9026	115262	9026	211987
Manipur	734	0	734	0	1390
Meghalaya	43	1	43	1	80
Mizoram	133	0	133	0	197
Nagaland	243	0	243	0	625
Odisha	6486	38	6486	38	9526
Puducherry	331	12	331	12	802
Punjab	4494	169	4494	169	6491
Rajasthan	16278	461			

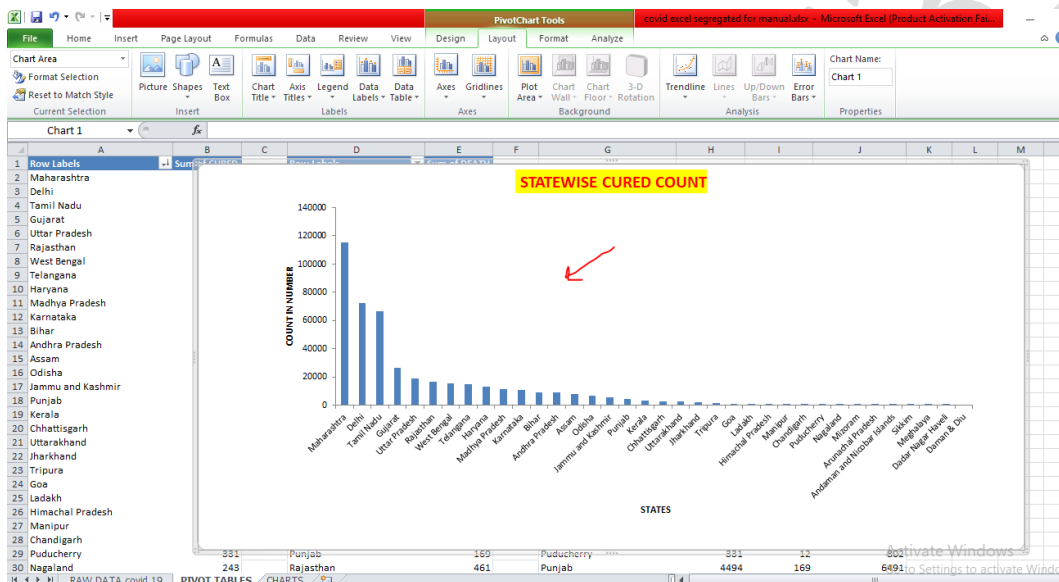
9) Now select particular pivot table data to put chart. Here I clicked on row label of first pivot and clicked on option. There we will get pivot chart option.

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The screenshot shows an Excel spreadsheet with a PivotTable. The PivotTable is located in the range A1:J10. The PivotTable has columns for State/UnionTerritory, Sum of Cured, Sum of Deaths, and Sum of Confirmed. The PivotTable Field List on the right shows the fields being used in the report.

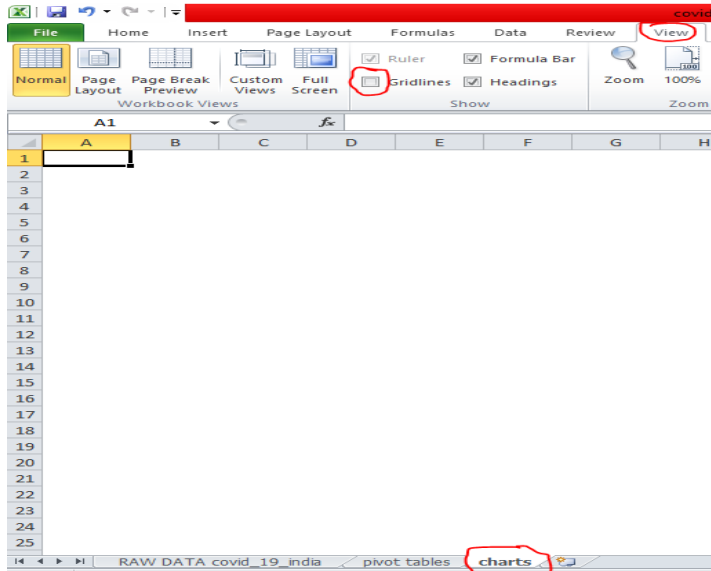
State/UnionTerritory	Sum of Cured	Sum of Deaths	Sum of Confirmed
Andaman and Nicobar Islands	2672	0	3
Andhra Pradesh	205693	6280	424
Arunachal Pradesh	994	13	13
Assam	113804	398	202
Bihar	203276	2133	322
Cases being reassigned to states	0	0	31
Chhattisgarh	14037	278	26
Chhattisgarh	47568	327	76
Dadar Nagar Haveli	14	0	0

10) Once you click on pivot chart it will ask you to select type of chart. As here we are having two variables to compare the data I will select column chart and click on ok.

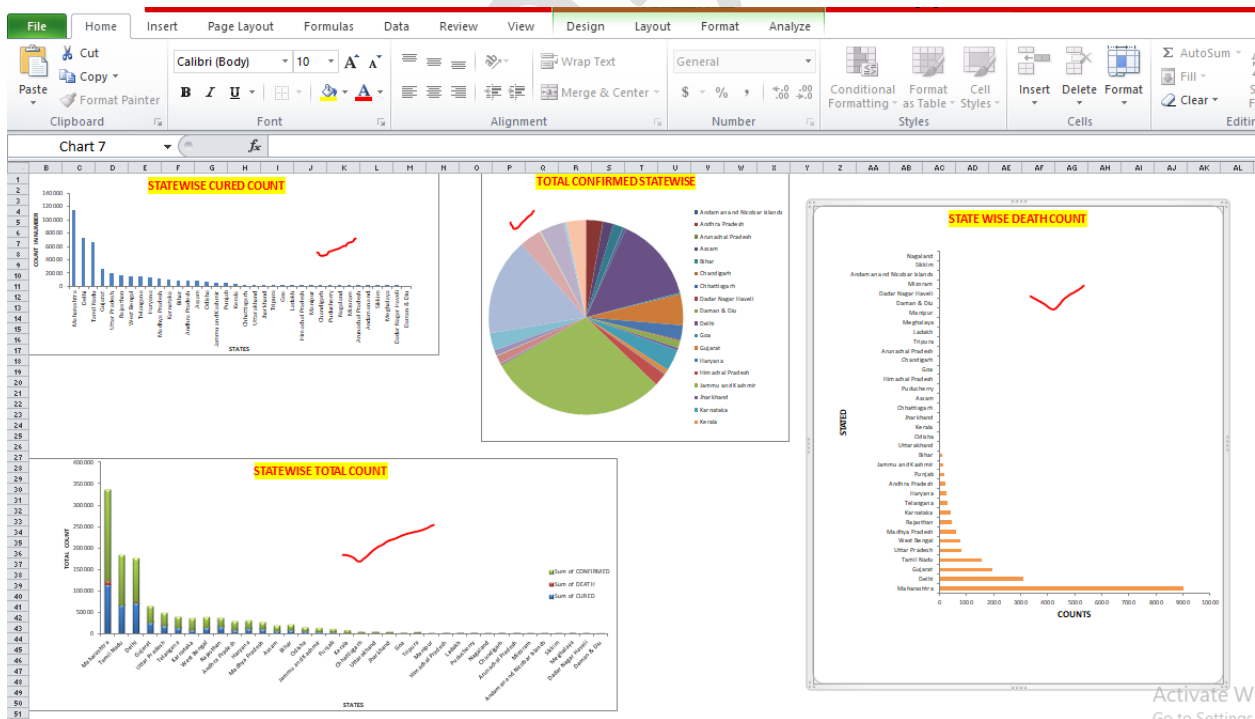


11) Go to new sheet (click on new sheet) and rename as charts. After renaming the sheet go to view and uncheck gridlines to make sheet plain.

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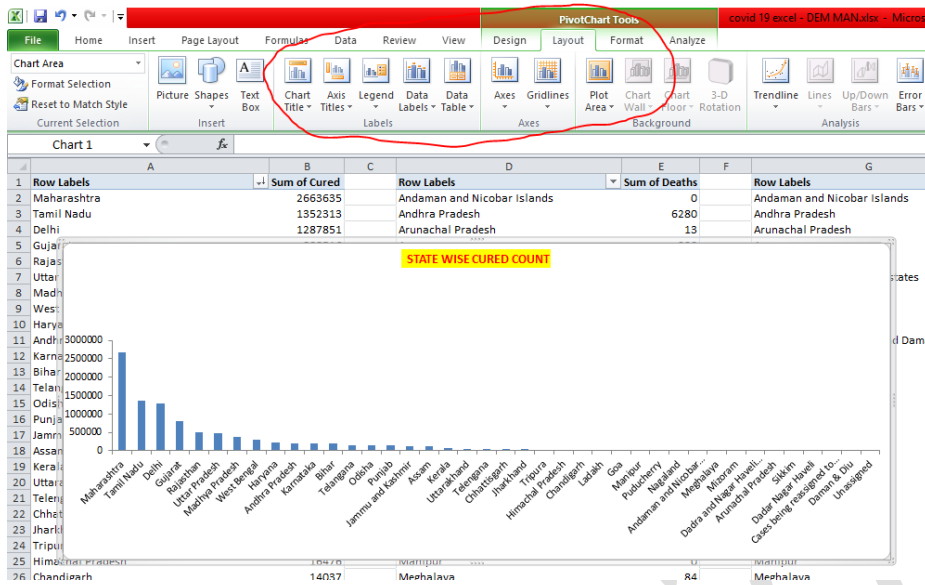


12) Repeat the procedure 9 and 10 to create required different charts, transfer all the charts drawn for pivot table using cut and paste option. Before doing it we should modify chart with effective visualization like color, axis title, chart title, data labeling etc.

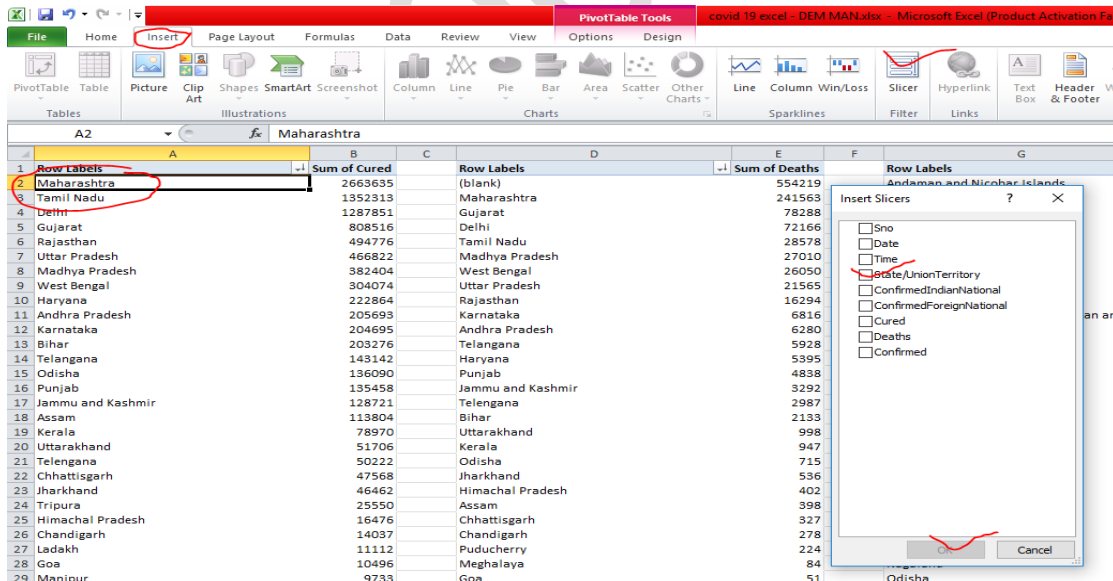


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13) These modifications can be done through layout option which will appear after clicking on chart.

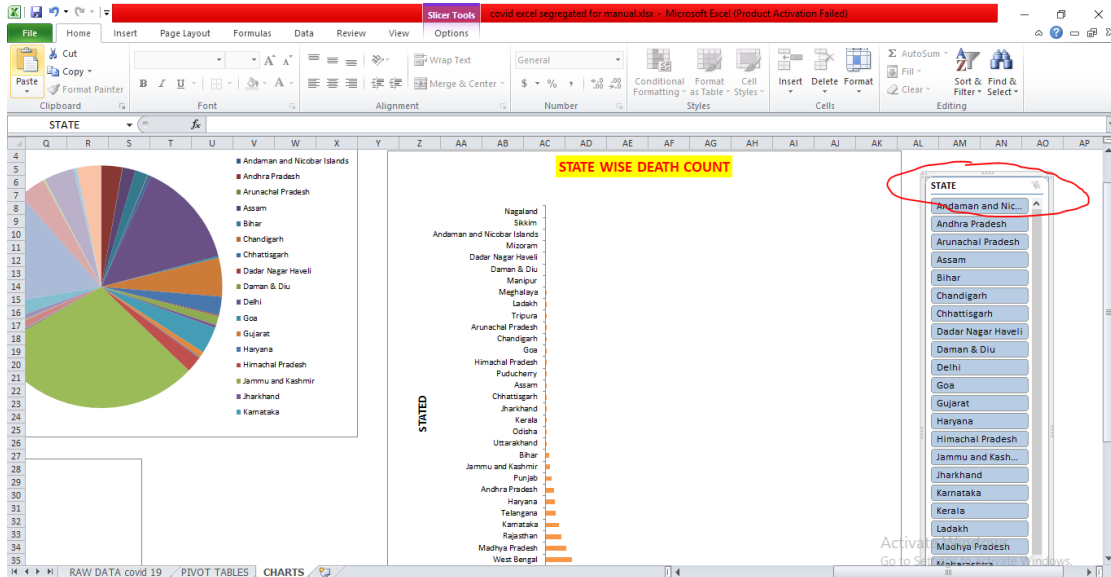


14) Once all charts are transferred to charts sheet after modification connect all pivot tables using slicer option. For that purpose go to pivot tables sheet then select any table > insert > click on slicer. Select state/union territory and click on ok.

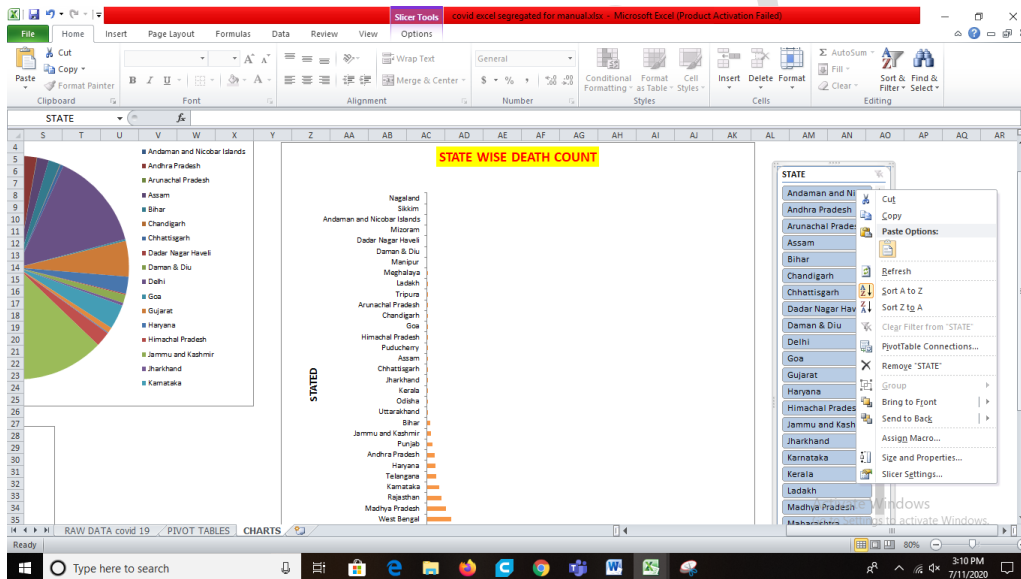


15) Now slicer has been added, cut and pastes that slicer to charts sheet.

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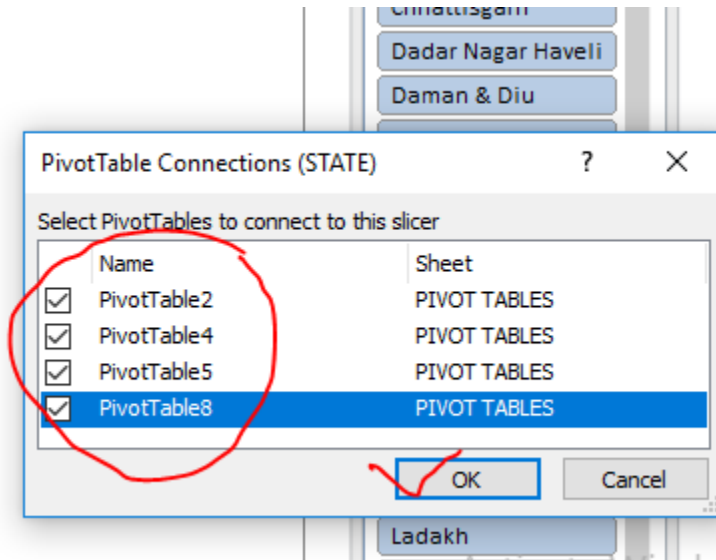


16) This slice is having link with only that particular chart now we need link this slicer to remaining chart by right clicking on that and use connect pivot tables option.

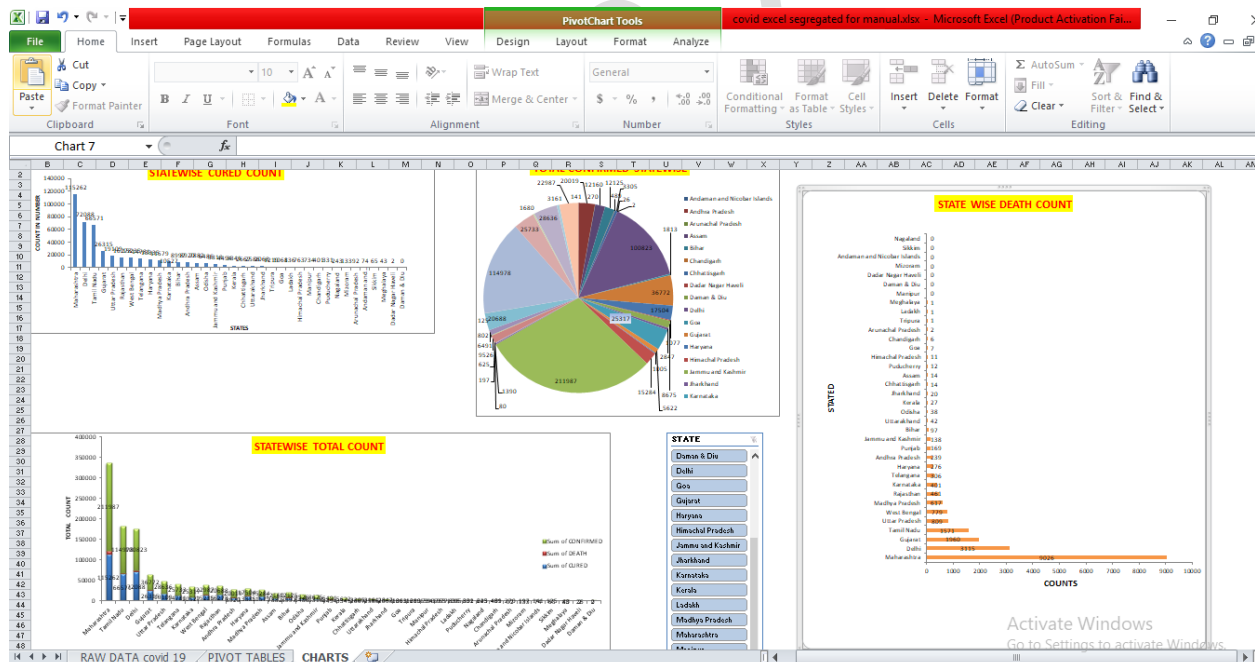


17) After clicking Pivot table connections, we will get a popup asking which all pivot table need to be connected? In this example we have selected all pivot tables and clicked ok as shown in figure.

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18) Now all pivot tables are connected to that slicer. We can now interact with simple dashboard.



RESULT: Simple dashboard using Microsoft Excel is created.

VIVA QUESTIONS:

1. What do you mean by data?

Data is raw facts and figures collected for analysis.

2. Briefly discuss importance of data visualization.

Data visualization helps understand patterns and make decisions easily.

3. List some of the data visualization tools.

Excel, Power BI, Tableau, Google Data Studio, Python.

4. What is a pivot table?

A pivot table summarizes and analyzes large datasets.

5. How will you connect pivot tables?

Pivot tables connect using slicers, timelines, and the same data source.

6. Is it possible to change dashboard chart style as per requirement?

Yes, chart styles, colors, and formats can be changed.

7. What is a dashboard?

A dashboard is a visual summary of key data in one view.

8. What is the difference between data science and data visualization?

Data science analyzes data; data visualization presents data visually.

9. List some of the charts you have gone through.

Bar, Line, Pie, Column, Scatter, and Area charts.

10. What are the prerequisites to start Excel dashboards?

Clean data, pivot tables, basic formulas, and chart knowledge